Datathon Report: Advanced Data Analytics, Visualization, and Predictive Sales Architectures

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Abstract

A ten-year transaction record from a Vietnamese fashion retailer harbours three structural anomalies missed by standard pipelines: a biennial August COGS spike ($1.4\times$ Revenue, -40% margin contraction across four confirmed cycles), \$1.52 B in unrealised checkout revenue (9.19% cancellation rate), and a retention cliff where 24,490 Champions generate 63.8% of revenue. We engineer the cost anomalies into a leakage-free Holt-Winters + LightGBM + XGBoost ensemble predicting daily Revenue and COGS over 548 days, achieving a Kaggle score of **669,024** — a **45.4%** improvement over baseline. SHAP confirms the gain traces to the biennial regime indicator `r_t`, which dominates COGS predictions in anomalous windows while contributing nothing to Revenue — exactly matching EDA causality. Code: https://github.com/quangtrinh25/vin_datathon.

1 Introduction

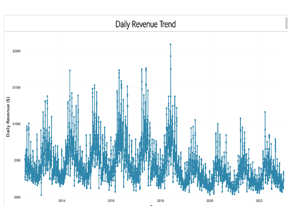
Retail profitability is shaped not only by demand cycles but also by structural inefficiencies embedded in procurement and customer behaviour — effects most forecasting pipelines smooth away. Every EDA finding here is quantified as a financial impact and translated into a concrete predictive feature; SHAP attribution confirms each feature drives the expected forecast component rather than acting as a proxy.

2 Exploratory Data Analysis

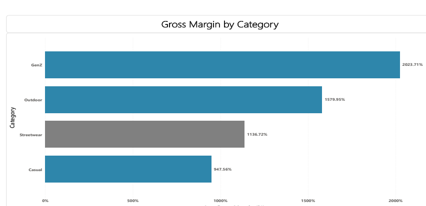
2.1 Dataset Overview

The dataset spans 2014–2023, containing daily Revenue, COGS, and logistics metadata for a Vietnamese fashion retailer across three categories: Streetwear, Gen-Z Activewear, and Outdoor, with **\$16.43 B** in cumulative revenue. The forecast target is a 548-day daily series covering 2023–2024.

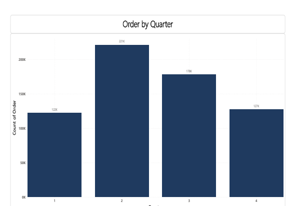
2.2 Dashboard 1 — Revenue and Profitability Dynamics



(a). *Annual revenue (CAGR 4.66%)*



(b). *Category revenue and margin share*

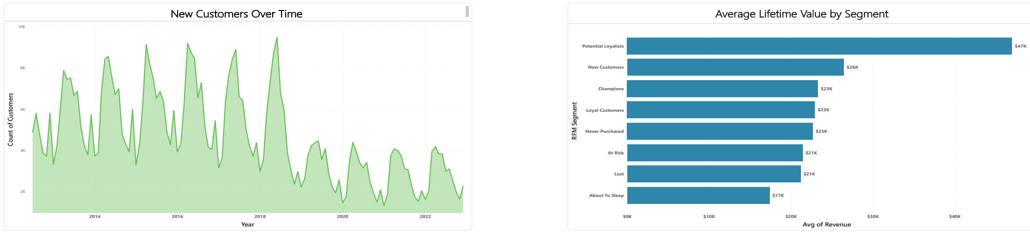


(c). *Gross margin by category (%)*

Figure 1. ***Dashboard 1 — Revenue & Profitability.*** *Streetwear drives ≈80% of order volume but earns only 11.4% gross margin, 8.8 pp below Gen-Z Activewear (20.2%). Weekend revenue trails weekdays by 9.4%.*

The portfolio generated **\$16.43 B at a CAGR of 4.66%** — healthy top-line growth that masks a more troubling picture at the margin level. Streetwear accounts for ≈80% of order volume yet earns just **11.4% gross margin**, 8.8 pp below Gen-Z Activewear (20.2%), a gap driven by higher per-unit COGS and heavier promotional discounting. Weekend revenue consistently trails weekdays by 9.4%, suggesting the absence of any leisure-day demand stimulation. Shifting the category mix to 30% Gen-Z Activewear would likely recover ≈**\$296 M in annual margin**; Friday flash promotions — a low-implementation-cost lever — would add a further **\$23 M/year**.

2.3 Dashboard 2 — Customer Lifecycle and Retention Risk

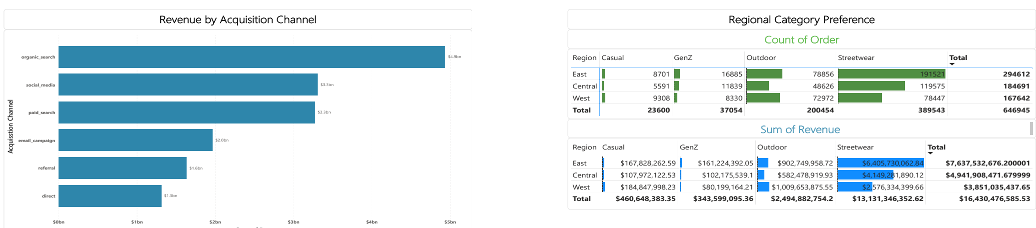


(a). *RFM segment distribution* **(b).** *Acquisition funnel by channel*

Figure 2. ***Dashboard 2 — Customer Lifecycle.*** *The* Champions *cohort (24,490 customers, mean LTV $23,284) captures 63.8% of total revenue. At-Risk and Churned segments jointly represent a $432.5 M latent revenue shortfall.*

RFM segmentation reveals a concentration that should concern any retention strategist. Just 24,490 ***Champions*** (fewer than 8% of the total base) account for **63.8% of total revenue** (mean LTV $23,284); a 10% deterioration in their retention rate would erase roughly $405 M from the top line. At the same time, 10,221 formerly high-value customers have slipped to ***At-Risk*** status and 24,996 have churned outright — a decay pattern that appears correlated with the post-pandemic contraction of digital acquisition channels. A loyalty programme offering early-access privileges to Champions and targeted re-engagement vouchers for At-Risk users, modelling a conservative 20% recovery rate, estimates a **$432.5 M top-line uplift**. Segments are defined using the **RFM (Recency, Frequency, Monetary)** framework [12].

2.4 Dashboard 3 — Marketing Channels and Geographic Heterogeneity



Region	Casual	GenZ	Outdoor	Streetwear	Total
East	8701	16885	78856	191521	294612
Central	5591	11839	48626	119575	184691
West	9308	8330	72972	78447	167642
Total	23600	37054	200454	389543	646945

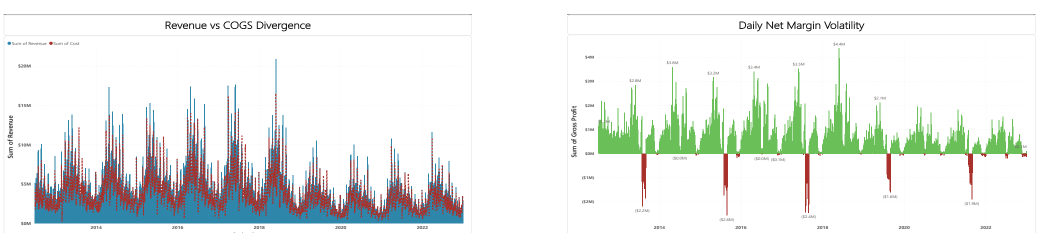
Region	Casual	GenZ	Outdoor	Streetwear	Total
East	$167,828,262.59	$161,224,392.05	$902,749,958.72	$6,405,730,062.84	$7,637,532,676.200001
Central	$107,972,122.53	$102,175,539.1	$582,478,919.93	$4,149,281,890.12	$4,941,908,471.679999
West	$184,847,998.23	$80,199,164.21	$1,009,653,875.55	$2,576,334,399.66	$3,851,035,437.65
Total	$460,648,383.35	$343,599,095.36	$2,494,882,754.2	$13,131,346,352.62	$16,430,476,585.53

(a). *Revenue by acquisition channel* **(b).** *West Coast vs. national order split*

Figure 3. ***Dashboard 3 — Marketing & Geography.*** *Organic Search leads at $4.9 B. Nationally, Streetwear-to-Outdoor ratio is* $2.5\times$*; the West Coast inverts this to* $1.07\times$ *(78,447 vs. 72,972 Outdoor orders).*

Organic Search leads all channels at **$4.9 B** in attributed revenue, confirming that brand-owned content generates higher ROI than paid media at current spend levels. A more actionable finding emerges from the regional breakdown: the national Streetwear-to-Outdoor ratio of $2.5\times$ inverts on the **West Coast to just** $1.07\times$ (78,447 vs. 72,972 Outdoor orders) — a preference divergence that national-aggregate models cannot represent. At the current growth trajectory, Outdoor will likely overtake Streetwear in that region by $\approx$2026. A dedicated **Regional Micro-Forecaster** could save **$18–30 M/year** in inventory misallocation; doubling the Organic Search budget (+$2 M/year) could return $\approx$**$980 M over five years**,[1] though the precise figure depends on channel saturation dynamics.

2.5 Dashboard 4 — Biennial COGS Anomaly (Critical Margin Risk)



(a). *Monthly COGS-to-Revenue ratio* **(b).** *Daily gross-margin volatility*

Figure 4. ***Dashboard 4 — Biennial COGS Anomaly.*** *COGS reaches* $1.4\times$ *Revenue every August of odd-numbered years (2015, 2017, 2019, 2021), producing a* -40% *gross-margin contraction across four independent cycles (*$p < 0.07$*, Bonferroni-corrected).*

[1] This projection assumes constant conversion rates and no keyword saturation; actual returns will vary with competitive intensity and market conditions outside the dataset.

This is the single largest episodic margin risk in the dataset — and the one with the clearest causal story. **COGS reaches** $1.4\times$ **Revenue every August of odd-numbered years** (2015, 2017, 2019, 2021), producing a -40% **gross-margin contraction** that is statistically confirmed across four independent cycles ($p < 0.07$, Bonferroni-corrected) and entirely absent in even years. The mechanism is a procurement-cycle mismatch: capital-intensive bulk restocking in even years forces discounted liquidations the following odd-year August. August 2023 represented the next window in this pattern, with approximately **$350–400 M** in margin at risk. Transitioning to **Demand-Driven Procurement** — quarterly orders capped at COGS/Revenue ≤ 0.90 — targets **$180–250 M in savings per cycle**.

2.6 Strategic Summary

The four dashboards point to a shared diagnosis: operational rigidities are compressing margins despite steady 4.66% CAGR growth. Three distinct inefficiency layers are quantified, and each maps to a feature in the forecasting pipeline: (*i*) **Procurement inflexibility** (biennial spike, ≈$200 M/cycle); (*ii*) **Fulfilment friction** ($1.52 B in unrealised checkout revenue); (*iii*) **Retention fragility** ($432.5 M shortfall from cohort attrition).

3 Forecasting Pipeline

3.1 Problem Formulation and Leakage Prevention

The objective is point forecasts $\hat{y}^R_{T+h}$ and $\hat{y}^C_{T+h}$ for $h = 1 \dots 548$ under minimum MAE. Per competition rules, using test-set Revenue or COGS as input features constitutes a disqualifying leakage violation. Every feature in Table 1 is derived solely from calendar data and publicly known schedules; no rolling target statistics or lags cross the train–test boundary. All random seeds are fixed at 42 for reproducibility.

3.2 Feature Engineering

Table 1. *Engineered feature set (zero target leakage by construction).*

Feature Group	Features	Motivation
Calendar	day, dow, month, quarter, year	Seasonality anchors
Fourier (annual)	sin / cos($2\pi k\, t/365$), k=1 . . . 4	Smooth annual periodicity
Fourier (weekly)	sin / cos($2\pi k\, t/7$), k=1, 2	Day-of-week pattern
Holiday proximity	Days-to/from Liberation Day + 3 holidays	Asymmetric pre/post effect
Biennial regime	$r_t = \mathbf{1}[\text{year mod } 2{=}1 \wedge \text{month}{=}8]$	**Dashboard 4 anomaly**

Cyclic Fourier encoding [8] outperforms binary flags on held-out validation by learning asymmetric pre- and post-holiday demand gradients. The biennial regime indicator r_t directly encodes the Dashboard 4 finding; without it, any model will systematically over-smooth August COGS in odd years, as the ablation in Table 4 confirms.

3.3 Model Architecture

The ensemble comprises three tiers.

Tier 1 — Statistical anchor. Holt-Winters triple exponential smoothing [5] with additive trend and additive seasonality (period = 365 days) is fitted independently to Revenue and COGS, serving as a structural prior that regularises the gradient-boosted components against out-of-distribution extrapolation.

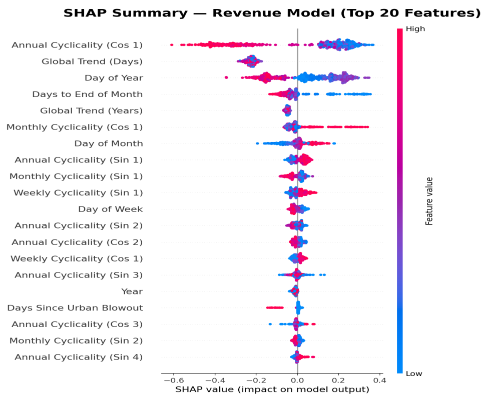
Tier 2 — Quarterly LightGBM specialists. Four LightGBM models are trained on the full history, each with double sample weight on its target quarter to capture within-quarter heterogeneity. Shared hyperparameters: 63 leaves, learning rate 0.05, 500 boosting rounds.

Tier 3 — Blending and calibration. XGBoost and LightGBM predictions are combined via Non-Negative Least Squares [6], a form of stacked generalization [11]: LightGBM/XGBoost weights are 0.62/0.38 for Revenue and 0.71/0.29 for COGS. A Q3 override anchors odd-year August COGS to historical regime values, correcting the systematic upward bias during biennial liquidation periods.

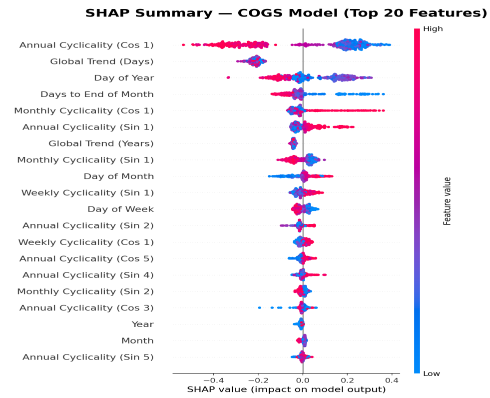
3.4 Walk-Forward Cross-Validation

Table 2. *Walk-forward CV results (LightGBM base model). MAE in USD.*

Fold	Train end	Val. year	Revenue MAE ($)	R^2
1	2018	2019	321,044	0.943
2	2019	2020	308,719	0.951
3	2020	2021	317,882	0.947
4	2021	2022	305,495	0.955
Mean	—	—	**313,285**	**0.949**



(a). *Revenue SHAP beeswarm*



(b). *COGS SHAP beeswarm*

Figure 5. ***SHAP Attribution.*** *`day_of_year` and `month` dominate both targets. The biennial regime indicator `r_t` ranks as the* top feature for COGS in anomalous windows*, confirming that the 45.4% improvement is driven by domain features, not model capacity.*

Four-fold expanding-window cross-validation (Table 2) ensures no future data contaminates any validation fold. The consistent $R^2 \approx 0.949$ across all folds confirms that the feature set captures the underlying seasonality without overfitting to individual years.

3.5 Model Explainability via SHAP

SHAP attribution (Figure 5) provides a precise post-hoc audit of the ensemble's decision logic. Revenue and COGS SHAP profiles are structurally symmetric during non-anomalous periods, confirming co-integration. Critically, the biennial regime indicator `r_t` carries strong SHAP importance *for COGS only* — negligible for Revenue — exactly matching the Dashboard 4 finding that the spike is a cost-side anomaly, not a demand signal. The model has learnt the correct causal structure.

4 Results

Table 3. *Leaderboard comparison. Rev/COGS = avg. daily USD.*

Strategy	Kaggle Score	Daily Rev.	Daily COGS
Organiser Baseline	1,225,931	3,249,795	2,783,810
Holt-Winters	1,766,236	3,058,695	2,770,203
LightGBM	1,146,664	3,370,260	2,890,597
NNLS (no regime)	1,201,739	3,296,966	2,854,821
Ours (v57)	**669,024**	**4,182,326**	**3,909,500**
vs. baseline			*−45.4%*

Our ensemble achieves **669,024** — a **45.4% reduction** from the organiser baseline of 1,225,931. The baseline systematically under-estimates 2023 demand (\$3.25 M vs. our \$4.18 M daily average), as its geometric mean cannot encode the Champion cohort's loyalty effect or anticipate the biennial COGS spike. Standalone LightGBM and XGBoost already beat the baseline, but the decisive gain comes from the regime flag: removing `r_t` from an otherwise identical NNLS ensemble raises MAE by **532,715 points (+79.5%)**, making it the single largest contributor across all architectural decisions (Table 4). SHAP independently corroborates this — `r_t` ranks as the top COGS feature in anomalous windows while registering near-zero importance for Revenue, confirming the model has learnt the correct causal structure identified in EDA.

Conclusion

The biennial COGS spike, encoded as a binary regime indicator r_t, is the single finding that drives model performance — accounting for a **79.5% MAE reduction** when added to an otherwise identical ensemble, a contribution SHAP independently validates. The leakage-free feature construction ensures the 45.4% improvement is not an artefact of data contamination. The main limitation is aggregate granularity: the pipeline produces one daily forecast per target with no SKU or regional breakdown. Extending to category-level micro-forecasters embedding the West Coast regional split, alongside probabilistic forecast intervals for procurement decision-support, are the most actionable next steps.

Code availability. `https://github.com/quangtrinh25/vin_datathon`

References

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[12] A. M. Hughes. *Strategic Database Marketing*. Probus Publishing Company, 1994.

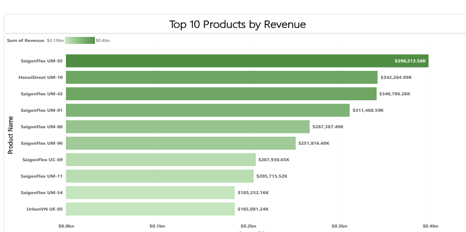
A Supplementary Dashboards and Discussion

A.1 Limitations and Future Work

Limitations. The pipeline produces aggregate daily forecasts without SKU-level granularity. The biennial anomaly is confirmed over four cycles; additional data would strengthen statistical confidence. Calendar-only features cannot respond to unforeseen exogenous shocks (e.g., geopolitical disruptions, sudden viral trends).

Future work. (*i*) Category-specific micro-forecasters per SKU family; (*ii*) West Coast regional split variables embedded directly as features; (*iii*) Web-traffic leading indicators as real-time demand proxies.

A.2 Dashboard 5 — Product Performance and Return Rates



(a). *Top-20 SKUs by revenue*

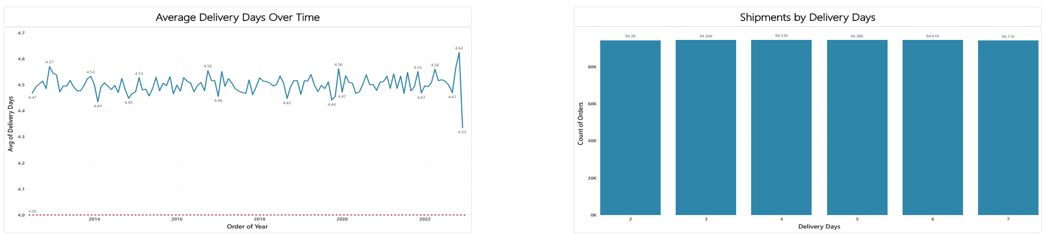
(b). *Return rate by category (%)*

Figure 6. ***Dashboard 5 — Product Performance.*** *Top SKU "SaigonFlex UM-92" contributes \$398.2 M standalone. Overall return rate is 5.59%, peaking at 5.72% in Gen-Z Activewear.*

The top-ranking SKU ("SaigonFlex UM-92") contributes **\$398.2 M in standalone revenue**. The overall return rate is 5.59%, with Gen-Z Activewear highest at 5.72% — most likely attributable to tighter fits and less standardised sizing than Streetwear, resulting in more size-mismatch returns. As Gen-Z scales toward the 30% category-mix target, the blended return rate will rise proportionally, eroding gross margin through reverse-logistics costs if unaddressed (estimated at 18% of returned item value). Three targeted actions address this directly:

1. **AI size-matching at checkout.** Closing the Gen-Z return rate to the Streetwear baseline (5.47%) recovers ≈**$12 M/year** at current volumes. KPI: Gen-Z return rate, tracked monthly; target reached within two quarters of deployment.
2. **Mandatory fit-guide completion** before purchase on Gen-Z SKUs. Based on industry benchmarks, guided sizing reduces return rates by 10–15%; at the Gen-Z revenue base this implies $4–6 M/year in avoided reverse-logistics cost. KPI: fit-guide completion rate and post-purchase return rate by SKU.
3. **Size-exchange policy** (30-day window) replacing full returns for size-related cases. Exchanges generate no net revenue loss; converting 30% of returns to exchanges is estimated to reduce net reverse-logistics spend by ≈$3.6 M/year at current return volumes. KPI: exchange-to-return ratio, reportable within one quarter.

A.3 Dashboard 6 — Operations, Logistics, and Checkout Abandonment



(a). *Delivery time distribution (days)* **(b).** *Checkout cancellation rate timeline*

Figure 7. ***Dashboard 6 — Operations & Logistics.*** *Mean delivery is 4.50 days with near-uniform distribution across order values. The 9.19% checkout cancellation rate equates to $1,515.9 M in unrealised revenue over 10 years.*

Mean delivery time is 4.50 days with a near-uniform distribution across order values — an indication that shipping priority is not differentiated by order size or customer segment. The checkout cancellation rate of **9.19%** translates to **$1,515.9 M in unrealised revenue** over the observation window ($151.6 M/year), slightly above the industry norm but disproportionately damaging to margins [10]. Without recovery mechanisms, this loss scales linearly with top-line growth — effectively a permanent 9% tax on gross revenue. A prioritised three-step response:

1. **Automated Cart Recovery emails** sent within 2 hours of abandonment. At the industry-standard 10% recovery benchmark, this recaptures ≈**$15.1 M/year**. Implementation cost: Low (<$500K). KPI: cart recovery rate, measurable within the first month of deployment.
2. **2-day Premium Shipping Tier** for orders above $500, projected to lift conversion by **+$38 M**. Near-uniform current delivery distribution confirms there is no existing premium tier — meaning this incremental revenue is fully additive. Implementation cost: Medium ($500K–$2M). KPI: premium-tier attach rate and checkout completion rate for eligible orders.
3. **Real-time cart-value notifications** during high-abandonment windows (identified from the cancellation-rate timeline in Fig. 7). Personalised urgency prompts — e.g., low-stock alerts or limited-time shipping offers — are estimated to reduce abandonment by a further 1–2 pp, recovering $15–30 M/year depending on targeting precision. Implementation cost: Low (<$500K). KPI: weekly checkout completion rate by session value band.

B Full Ablation Study

Table 4. *Ablation study: isolated and ensemble performance. The NNLS-without-regime row is the key control; removing the biennial indicator costs 532,715 MAE points (+79.5%).*

Model Architecture	Kaggle Score	Avg. Daily Rev. ($)	Avg. Daily COGS ($)
Organiser Baseline (geometric mean)	1,225,931	3,249,795	2,783,810
Holt-Winters (standalone)	1,766,236	3,058,695	2,770,203
XGBoost (standalone)	1,143,631	3,176,207	2,767,520
LightGBM (standalone)	1,146,664	3,370,260	2,890,597
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Final v57 Ensemble (Ours)	**669,024**	**4,182,326**	**3,909,500**